

CURRICULUM VITAE

Name: Reijul Sachdev

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Date of Birth: 27 October 1993

Current Occupation: PhD scholar at IIIT-Bangalore

Work Experience:

- Worked as a Teaching Assistant for a semester at IIIT-B, teaching Statistics for Data Science
- Worked as an Associate at Round Glass H2O Pvt. Ltd.
- Completed a 3-month internship as a data scientist at Round Glass
- Took admission to IITM for a PhD in cryptography
- Research Associate in the Multimodal Perception Lab (MPL) and the Machine Intelligence and Robotics Lab (MINRO) at the International Institute of Information Technology, Bangalore for a year

Academic Achievements:

- Invited to attend the first Indian Conference on Information Systems (InCIS) at IIMC in March 2025
- Invited to submit a paper titled ‘Modeling and Analysis of Whistle-blowing: An aid to Data Protection Regulation in India’ to **Special Issue of the *Pacific Asia Journal of the Association for Information Systems (PAJAIS)*** (currently under review)
- Awarded the Sir M. Visvesvaraya Scholarship Award (Male) and corresponding Gold Medal for the highest CGPA in all streams for 2019 by the International Institute of Information Technology, Bangalore (IIIT-B)
- Wrote and successfully defended Master’s thesis titled ‘An Architecture for Sensex Prediction and Black Swan Detection using Neural Networks and Wavelet Transforms’
- Batch topper with an overall CGPA of 3.98/4
- Completed 3-month internship at the MIT Media Lab with the Human Dynamics and Changing Places Groups
- Granted admission to St. Stephen's College, Delhi for B.Sc. Physics (Hons.), but refused it to get admitted to the 5-year Integrated M.Tech. program at IIIT-B
- Eligible for INSPIRE scholarship (awarded by CBSE to students pursuing pure science) based on grade 12 result

- School topper in Grade 12 of National Public School, Koramangala (CBSE) with 96.8%
- Only student to score above 90% in all subjects from Greenwood High in 2010 ICSE batch

Projects and Research Work:

1. Conducting research for a paper on economics of privacy and game theory as well as quantum error correcting codes for security and privacy.
2. Studied the theory behind GANs and gave several lectures on different types of GANs to the Multimodal Perception Lab. Also applied this theory to a project for semantic face editing using latent space manipulation, in collaboration with the University of Lausanne, Switzerland.
3. Completed thesis work using a combination of sentiment analysis, long short-term memory networks (LSTMs) and Haar wavelet transforms for the purpose of predicting changes in the direction as well as the magnitude of the Sensex.
4. Worked on mobility and social network modelling from wifi and CDR data for the Andorra Project in the MIT Media Lab with Professors Sandy Pentland and Kent Larson.
5. Completed a Research Elective on Neural Turing Machines (NTMs).
6. Completed an Elective on wavelets and other signal processing techniques for graphs and other non-Euclidean spaces (implemented in the context of the Andorra project).
7. Completed an Elective on Blockchains and Cryptocurrencies.
8. Built a convolutional neural network (CNN) from scratch in MATLAB for pattern recognition (in this specific case, traffic sign recognition) as part of a Project Elective.
9. Solved AMM (American Mathematical Monthly) problems 11685 and 11734 and was active member of the IIIT-B Maths Club.
10. Presented a talk on Neural Networks as part of the Multimodal Perception Lab in IIIT-B.
11. Completed an Elective in Digital Image Processing with projects involving Canny Edge detection, Hough detectors, etc.

12. Completed an Elective on Fundamentals of Cryptography and Multi-Party Communication.
13. Completed both pitch extraction project for various musical instruments as well as speaker localisation projects as part of the DSP (Digital Signal Processing) course.
14. Worked on part (along with Professor V. Muralidharan) of railway scheduling project for the Government of India. To this end, designed a simplified railway scheduling system in Python.
15. Attended a workshop on Differential Equations at NIT Bhopal, held by the faculty of IIT-B, IIT-K and LNMIT-Jaipur.
16. Designed and implemented a communication channel from start to finish, including source coding (LZW), channel coding (Golay and Viterbi) and modulation stages.
17. Developed a simulation of an e-commerce portal using JDBC.
18. Designed a simple OS shell using the Pintos kernel.
19. Developed a version of Othello (Reversi) for networked play in C (LINUX environment).
20. Programmed and used iRobots to develop a variant of Pacman with user and system-controlled iRobots, using playerstage.
21. Developed Morse code transmitter and receiver using various physical signals.
22. Implemented various types of schedulers in C for OS, including FCFS, RR, HRRN, etc.
23. Designed a bit flipping technique for saving power wasted in memory storage in DRAM circuits and tested in SimpleScalar.
24. Heart rate variability calculation using Audacity.
25. Participated in IEEE Xtreme and ACM ICPC programming contests.
26. Designed video game Pacman in grade 12 as part of a school project (using C++).

Technical Skills:

Programming proficiency in C, C++, Java, Python, MATLAB, GNU Octave, QBASIC,

OCaml and MySQL. Familiar with Keras and TensorFlow 2.0. Competence in using Audacity, Dia, LaTeX and Microsoft Dynamics NAV 2013.

Co-curricular Activities:

- Published a hilarious, psychological thriller titled “Wildcat” with Notion Press in April 2023
- Published a detective novel titled “Hai-Clue” with Notion Press in April 2021.
- Won the Best Overall Work Award from Elite Book Awards.
- ‘Hai-Clue’ won the Sahityakosh Samman award from NE8x, a platform for promoting literary talent.
- Won the Bharatiya Youth Face of India award and was chosen as one of the top 100 most inspirational people in Asia.
- Won the Pride of India award organised by cherrybookawards, an international book agency.
- ‘Hai-Clue’ nominated for the Tagore Commemorative Award for Literary Excellence.
- Featured in celebrity magazine ‘Inspirant’.
- Won first place in ‘Three Inspiring Poems by Contemporary Poets’ and had poem published in Hindustan Metro, Google News, Flipboard, Dailybeat and Punjab Metro.
- ‘Hai-Clue’ received a rating of 4.77/5 on Goodreads and 4.7/5 on Amazon.in.
- ‘Hai-Clue’ featured on NE8x’s reading list for the week and book of the day in August.
- Published a book of poetry titled “Emerald Blades” with Leadstart Publishing in May 2016. The book was well-received by critics with poetry reading sessions held at the British Council and the Bangalore Literature Festival. 'Emerald Blades' also won the 'Write, Publish, Publicize' contest held by the Bangalore Poetry Festival.
- ‘Emerald Blades’ sold nearly 200 copies and received a rating of 4.8/5 on Goodreads and 4.7/5 on Amazon.in.
- One of the winners of the Friedrich Neumann Foundation's Liberal Essay Writing Competition on "What it means to be a liberal in India?"
- Had a poem published in a poetry anthology called “Po’try”, after winning the poem-writing competition held by publishing house RainDrops Company Bangalore.
- One of 12 winners of the I Lead India (a Times of India and TIMES NOW initiative) programme and appeared on TIMES NOW Youth Parliament TV debate as a consequence.
- Winner (of both Best Speaker and Best Team awards) in the Frank Anthony Memorial All-India Interschool Debate (for all ICSE schools) competition.
- Winner (of both Best Speaker and Best Team awards) of the regional round of Debating Matters India (DMI) debate (held by British Council).
- Active member of DMI alumni association and later chaired debates on behalf of British Council.

- Member and active participant of IIIT-B Maths Club.
- Participated in IEEE Xtreme programming contest and emerged first amongst all college teams.
- Participated and won prizes in numerous inter-college fests, including IISc's Open Day, Mount Carmel College's Cassiopoeia, SIMC's Prayudh, etc.
- Science Club President in NPS-K (grade 12) and organised 'Khoj', the Science fest of NPS-K.
- School Vice-Captain in Greenwood High.
- Participated in numerous inter and intra-school events, including theatre, creative writing, programming, mock stock, elocution, etc.